



Curve Fitting



Hyperelastic



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Parameters in hyperelastic model curve fitting



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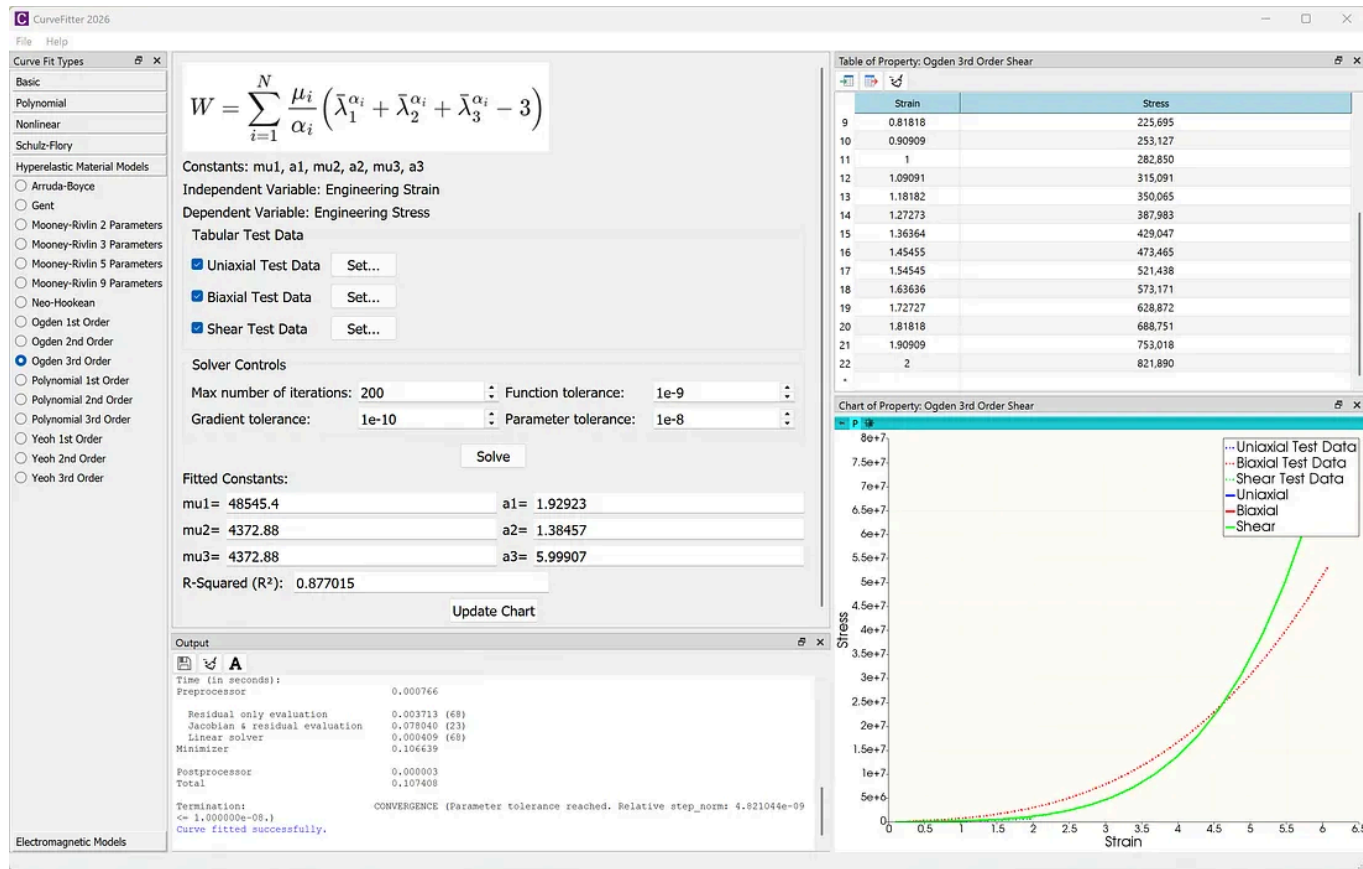
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Many tasks in finite element analysis require curve fitting. Especially in materials engineering, when only experimental test data is available, engineers must perform curve fitting to determine the parameters in a given numerical model. Common types of material curve fitting include hyperelastic, plastic, viscous, core loss materials, and more.



Unlike other types of curve fitting, curve fitting for finite element material models requires both a small error and physically valid fitted parameters; furthermore, these parameters must be conducive to convergence in subsequent nonlinear finite element computation. As a result, there are high standards placed on the algorithms and details of parameter fitting.

Rarely do articles discuss the parameters, initial values, and reasonable bounds for hyperelastic model parameters despite their importance when

implementing hyperelastic model curve fitting and verifying the rationality of fitted parameters. From a development perspective, this article discusses how to better perform curve fitting for hyperelastic material models, with a particular focus on setting parameter bounds and initial values, as well as identifying the characteristics of reasonable parameters.

Due to the large number of hyperelastic models and order-dependent variation, this article describes the computational details of each hyperelastic model in separate sections to avoid confusion.

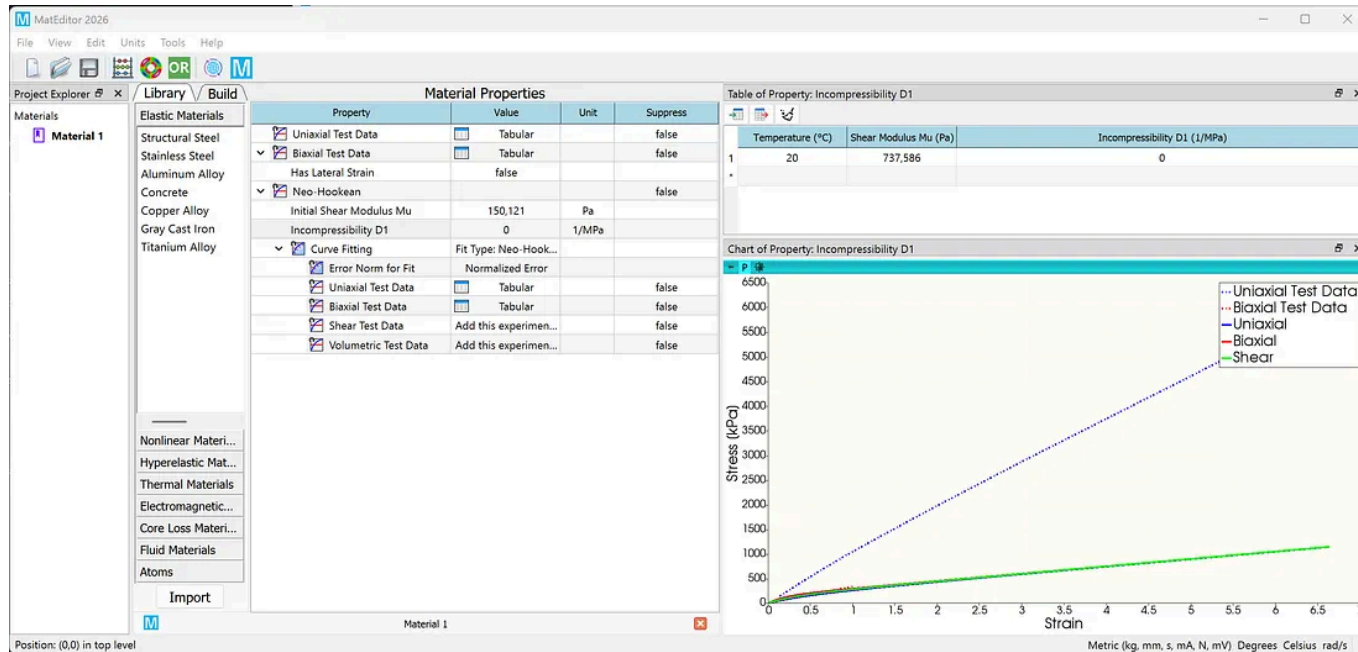
1. neo-Hookean

The neo-Hookean model is first because it is one of the most simple and classic hyperelastic models. Many other models can be simplified to the neo-Hookean model under specific parameter conditions. Its strain energy density function W is expressed as:

$$W = \frac{\mu}{2} (\bar{I}_1 - 3)$$

where μ is the shear modulus. During curve fitting computation, the shear modulus must be greater than zero, i.e., $\mu > 0$. The initial value is set to μ_0

$\sigma = E/3$, where Young's modulus E can be obtained from the slope of the stress-strain curve near the origin (strain < 5%).



To check for physical validity after parameter fitting, you can refer below for the approximate correspondence between common hyperelastic materials and μ :

Material Type	Shear Modulus μ (MPa)
Extra-soft Rubber	0.3 ~ 0.5
Medium-hardness Rubber	1.0 ~ 1.5

Hard Rubber	3.0 ~ 5.0
Biological Soft Tissue	0.01 ~ 0.1

A fitted parameter μ that differs significantly from the uniaxial tension data indicates that the material may exhibit a strong second strain invariant effect, in which the Mooney-Rivlin or other suitable models should be used instead.

2. Mooney-Rivlin

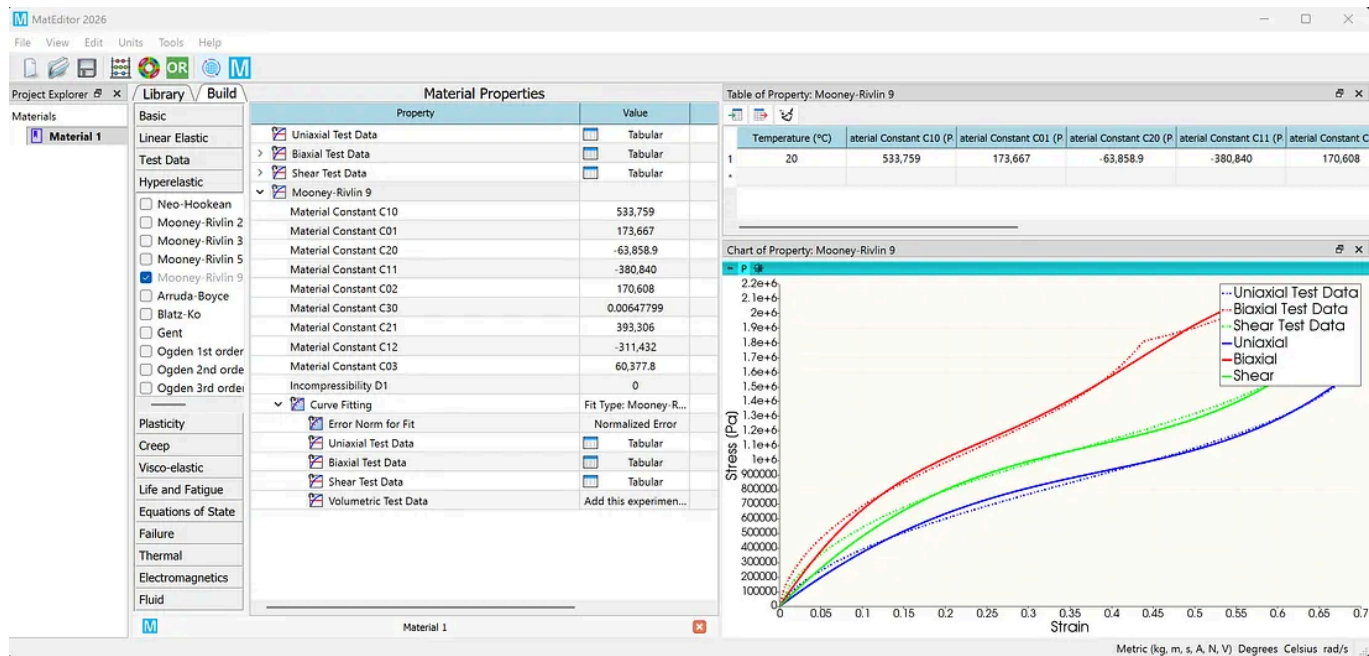
The Mooney-Rivlin model is another one of the most commonly used hyperelastic models in engineering analysis. It is a direct extension of the neo-Hookean model and is characterized by its simple and flexible form. If the neo-Hookean is considered the entry-level model in hyperelasticity, then the Mooney-Rivlin would be the standard model for handling small-to-moderate deformations (100%-200% strain). Its simple mathematical form and linear dependence on material constants allows it to converge easily in Finite Element Analysis (FEA) calculations.

Depending on the order, the Mooney-Rivlin model has several different forms with varying amounts of parameters, the most complex being the 9-parameter model. This section uses the 9-parameter Mooney-Rivlin model as an example to discuss parameter ranges and initial value settings. Its strain

energy density function W includes all combinations from linear to cubic terms:

$$\begin{aligned} W = & C_{10} (\bar{I}_1 - 3) + C_{01} (\bar{I}_2 - 3) + C_{20} (\bar{I}_1 - 3)^2 \\ & + C_{11} (\bar{I}_1 - 3) (\bar{I}_2 - 3) + C_{02} (\bar{I}_2 - 3)^2 + C_{30} (\bar{I}_1 - 3)^3 \\ & + C_{21} (\bar{I}_1 - 3)^2 (\bar{I}_2 - 3) + C_{12} (\bar{I}_1 - 3) (\bar{I}_2 - 3)^2 + C_{03} (\bar{I}_2 - 3)^3 \end{aligned}$$

Physically, the shear modulus $\mu > 0$. Therefore, $\mu = 2(C_{10} + C_{01}) > 0$. The most robust constraint for practical calculations is $C_{10} > 0$ and $C_{01} \geq 0$. In many rubber materials, C_{10} is the dominant term, and C_{01} is relatively small; C_{11} is also usually positive. The third-order parameter $C_{30} \geq 0$. C_{03} , C_{21} , C_{12} can capture the sharp stiffening at extremely large deformations (strain > 400%) and are generally 3–4 orders of magnitude smaller than the first-order parameters.



Regarding initial conditions, $C10=0.9 \times (\mu_0/2)$, $C01=0.1 \times (\mu_0/2)$, with the initial shear modulus $\mu_0=E/3$. The initial values for $C11$, $C20$, $C02$, $C30$, $C21$ are often chosen to be small, typically 0.1% of $C10$. The initial values for $C12$ and $C03$ can be set to zero. For most synthetic rubbers (e.g., NBR, EPDM, Neoprene), the parameters generally follow the order-of-magnitude rules below:

Order Classification	Parameter Symbol	Typical Value Range (MPa)
1st-order (Linear)	C10,C01	0.1 ~ 2.0

2nd-order (Transition)	C20,C11,C02	-0.2 ~ 0.2
3rd-order (Stiffening)	C30,C21,C12,C03	-0.05 ~ 0.1

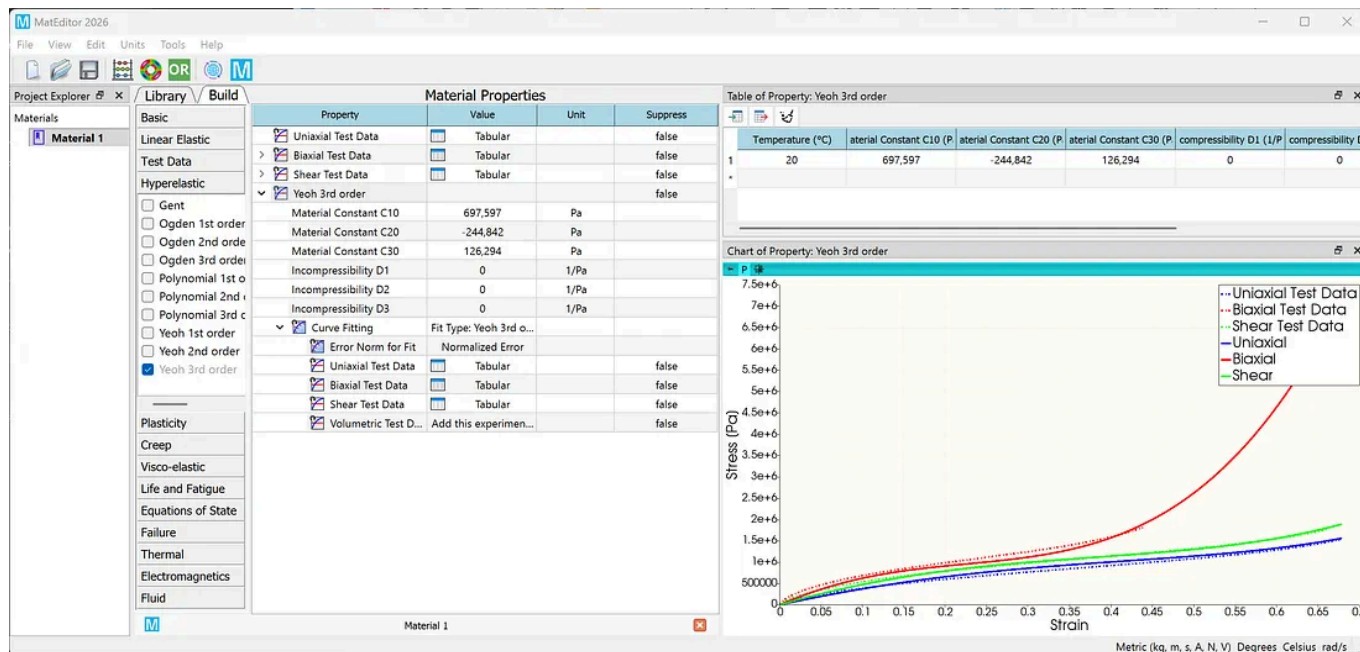
The Mooney-Rivlin model has its limitations. When rubber is stretched to its limit (molecular chains fully extended), the stress rises sharply (the “upturn” phenomenon). The Mooney-Rivlin model cannot account for this stiffening at high strains; the Ogden or Gent model is recommended when simulating extremely large deformations (strain > 300%). If the experimental data is not comprehensive, a more robust model with fewer parameters is advised, such as the Yeoh model or the 5-parameter Mooney-Rivlin model.

3. Yeoh

The Yeoh hyperelastic model is known for its simplicity and stability. It discards the second invariant I2 (terms like C01), which is highly susceptible to experimental error, retaining only the first invariant I1. Therefore, it is more robust than the Mooney-Rivlin model when fitting large-deformation data. The strain energy density function W is as follows:

$$W = \sum_{i=1}^N c_{i0} (\bar{I}_1 - 3)^i$$

Common Yeoh models are 1st to 3rd-order, with 1 to 3 parameters respectively. During fitting, $C10 > 0$ is required since the material loses shear resistance and crashes the simulation immediately if $C10 \leq 0$. $C20$ is usually negative; its magnitude is not too large and can be constrained to be greater than 1% of $C10$. $C30$ is positive and is used to capture the stiffening at large strain.



The initial value settings are relatively simple: the initial value of $C10$ is $E/6$, where Young's modulus E is the slope of the test data at small deformations. $C20 = -0.1 \times C10$, $C30 = 0.01 \times C10$.

The parameters of the Yeoh model for common materials are as follows:

Material Type/Hardness	C10 (MPa)	C20 (MPa)	C30 (MPa)
Ultra-soft Silicone/Gel	0.01 ~ 0.1	-0.001 ~ -0.01	0.0001 ~ 0.01
Natural Rubber (40A-50A)	0.2 ~ 0.5	-0.02 ~ -0.1	0.01 ~ 0.05
Industrial-grade Rubber (60A-70A)	0.6 ~ 1.2	-0.1 ~ -0.5	0.1 ~ 0.5
High-hardness Elastomer (80A+)	1.5 ~ 5.0	-0.5 ~ -2.0	0.5 ~ 2.0

4. Polynomial

The Polynomial hyperelastic model is also based on the power series expansion of the strain energy density function W with respect to the invariants I_1 and I_2 . It is one of the most widely used frameworks in hyperelasticity theory, and many models (Mooney-Rivlin, Yeoh, Neo-Hookean) are essentially its special cases. The general strain energy density function W is expressed as:

$$W = \sum_{i+j=1}^N c_{ij} (\bar{I}_1 - 3)^i (\bar{I}_2 - 3)^j$$

where N is the order of the model. Common polynomial models are 1st to 3rd-order, corresponding to 2, 5, and 9 parameters. Like the Mooney-Rivlin model, during fitting, it is necessary for $C_{10} > 0$ and $C_{10} > |C_{01}|$. C_{30} is also usually required to be positive. Generally, the higher the order, the smaller the magnitude of the values. The reasonable parameter ranges for hyperelastic materials are shown below.

Parameter Category	Parameter Item	Typical Range (MPa)	Physical B
Base Modulus	C_{10}	0.1 ~ 0.6	Dominates
Shear Correction	C_{01}	0.01 ~ 0.1	Corrects s
Mid-range Adjustment	C_{20}	-0.05 ~ 0.05	Controls s
Large-deformation Stiffening	C_{30}	0.001 ~ 0.01	Controls t
Coupling Terms	C_{11}, C_{02}	-0.01 ~ 0.01	Cross-effe

For initial value settings, set $C_{10} = E/6$, $C_{20} = -0.1 \times C_{10}$, $C_{30} = 0.01 \times C_{10}$ or smaller. Although the value of C_{30} is small, its high power amplifies its impact on the end of the curve.

5. Arruda-Boyce

The Arruda-Boyce model is unique in hyperelastic material simulation, and it is constructed based on statistical mechanics (molecular chain network theory). The strain energy density function W is expressed as:

$$\begin{aligned}
W = & \mu \left[\frac{1}{2} (\bar{I}_1 - 3) + \frac{1}{20\lambda_m^2} (\bar{I}_1^2 - 9) + \frac{11}{1050\lambda_m^4} (\bar{I}_1^3 - 27) \right. \\
& \left. + \frac{19}{7000\lambda_m^6} (\bar{I}_1^4 - 81) + \frac{519}{673750\lambda_m^8} (\bar{I}_1^5 - 243) \right]
\end{aligned}$$

A key feature of this model is its low requirement for test data, and uniaxial data alone is sufficient to start. Unlike polynomial models that require multi-axial data to determine parameters, the Arruda-Boyce model's parameters have clear physical definitions, in which relatively accurate μ and λ_m can usually be obtained from the high-quality uniaxial tension data.

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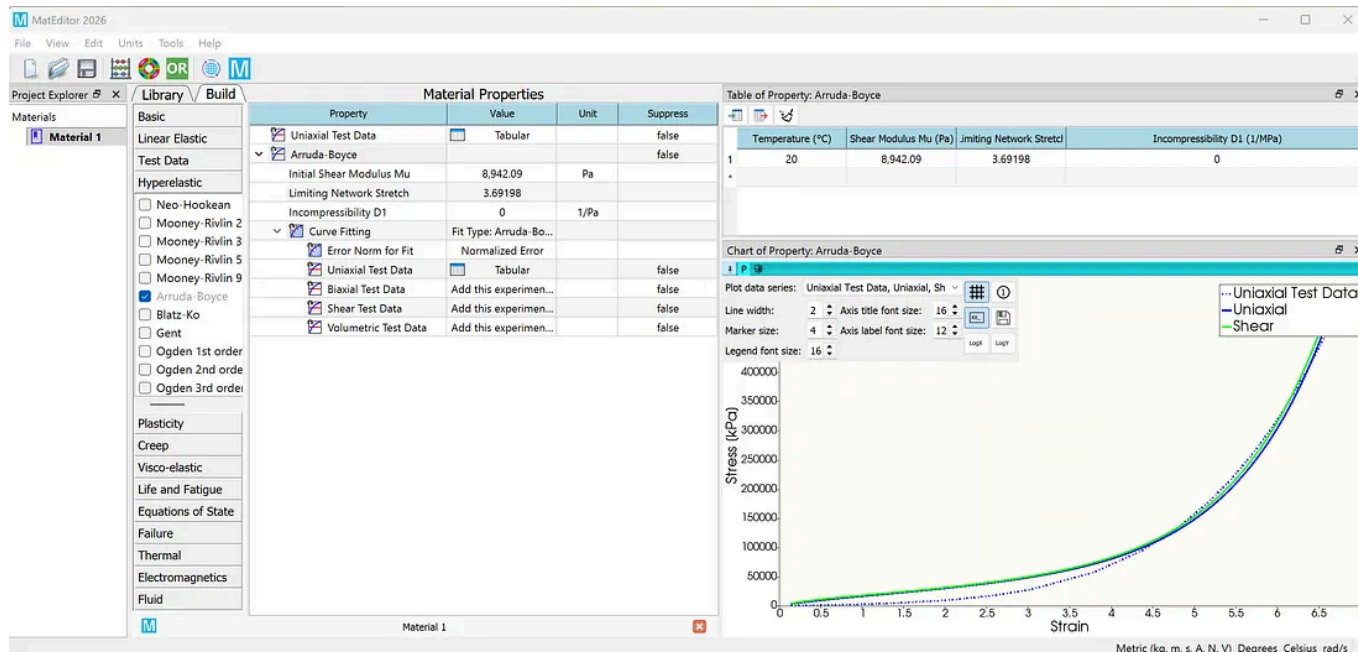
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The value range for the initial shear modulus μ is straightforward; it must be greater than 0 and lie between 0.1~5 MPa, with common values for rubber materials ranging from 0.5~1.5 MPa. λ_m must be greater than and typically limited to [1.1, 20]. The model becomes extremely stiff when λ_m is too close

to 1, leading to highly unstable numerical calculations. For most industrial rubbers, setting λm between 3.0 and 7.0 usually covers most working conditions.



Initial value determination is also straightforward: set $\mu = E/3$. $\lambda m = 5$ is a robust starting point.

Typically, the Arruda-Boyce parameters correspond to rubber materials as follows:

Material	μ (MPa)	λ m
Soft Rubber (Shore 30A - 40A)	0.3 ~ 0.6	5.0 ~ 8.0
Medium-hardness (Shore 50A - 60A)	1.0 ~ 2.5	3.0 ~ 5.0
Hard Rubber (Shore 70A - 80A)	3.0 ~ 8.0	1.5 ~ 2.5

6. Blatz-Ko

The Blatz-Ko model is a simple model used to simulate foams and porous materials. The most commonly used form contains only one parameter, the shear modulus μ .

$$W = \frac{\mu}{2} \left(\frac{I_2}{I_3} + 2\sqrt{I_3} - 5 \right)$$

Like other hyperelastic models, the shear modulus must be greater than zero. Poisson's ratio ν is fixed as a theoretical property of this model is that its effective Poisson's ratio is locked at approximately 0.25. This means the material is assumed to shrink significantly in volume when compressed, rather than expanding sideways like ordinary rubber. Do not use the Blatz-Ko model for solid rubber (Poisson's ratio near 0.5, incompressible).

As long as the Blatz-Ko model ensures that the shear modulus μ is positive ($\mu > 0$) and the material is indeed a compressible foam, fitting is usually straightforward.

Since Poisson's ratio ν is often fixed at 0.25, according to the linear elastic conversion relationship: $\mu = E/[2(1+\nu)] = E/2.5 = 0.4E$. The initial value can be set to $0.4E$.

Typically, the shear modulus corresponds to the materials as follows:

Material Type	μ Value Range (MPa)	Description
Ultra-light Polyurethane Foam	0.05 ~ 0.5	Extremely low density, high resilience
Medium-density Foamed Rubber	0.5 ~ 2.0	Common for shock pads and vibration isolation
High-hardness Closed-cell Foam	2.0 ~ 10.0	Structural support foam, high load capacity
Solid Compressible Rubber	10.0+	Special synthetic rubbers for high-pressure applications

7. Gent

The Gent model is a physically based hyperelastic model, similar in behavior to the Arruda-Boyce model. Its greatest advantage is its simple form, which can effectively capture the nonlinear behavior of polymer chains reaching their limiting extension length using only two constants. The mathematical

form of the Gent model is a logarithmic function, and the computational cost for curve fitting is lower than that of the Arruda-Boyce model.

The Gent model has only two core parameters: μ (initial shear modulus) and J_m (limiting invariant). Its strain energy density function is defined as:

$$W = -\frac{\mu J_m}{2} \ln \left(1 - \frac{\bar{I}_1 - 3}{J_m} \right)$$

The parameter μ must be greater than 0; J_m must be greater than the maximum measured value of $I_1 - 3$. If J_m is too small, causing $(I_1 - 3)/J_m \geq 1$, the logarithm becomes undefined, so the calculation will fail.

The initial value of the shear modulus μ is simply determined by $\mu = E/3$. Mathematically, J_m defines the upper limit for $I_1 - 3$, so its initial value must be slightly larger than the invariant value corresponding to the maximum deformation observed in the experiment. To determine this, find the maximum elongation ratio $\lambda_{\max}$ in the experimental data and calculate the corresponding I_1 . Note that the formula for I_1 varies slightly depending on the type of experiment. The initial value of J_m is set to $1.2 \times (I_{1_max} - 3)$.

The empirical parameters for materials of different hardness are as follows:

Material Hardness/Type	Shear Modulus μ (MPa)	Limiting Constant J_m	D
Ultra-soft Gel/Soft Tissue	0.01 ~ 0.15	0 ~ 150	E
Natural Rubber (40A)	0.3 ~ 0.6	30 ~ 80	E
Industrial Rubber (60A)	1.0 ~ 2.0	10 ~ 25	M
High-hardness Elastomer (80A)	3.0 ~ 7.0	2 ~ 8	S

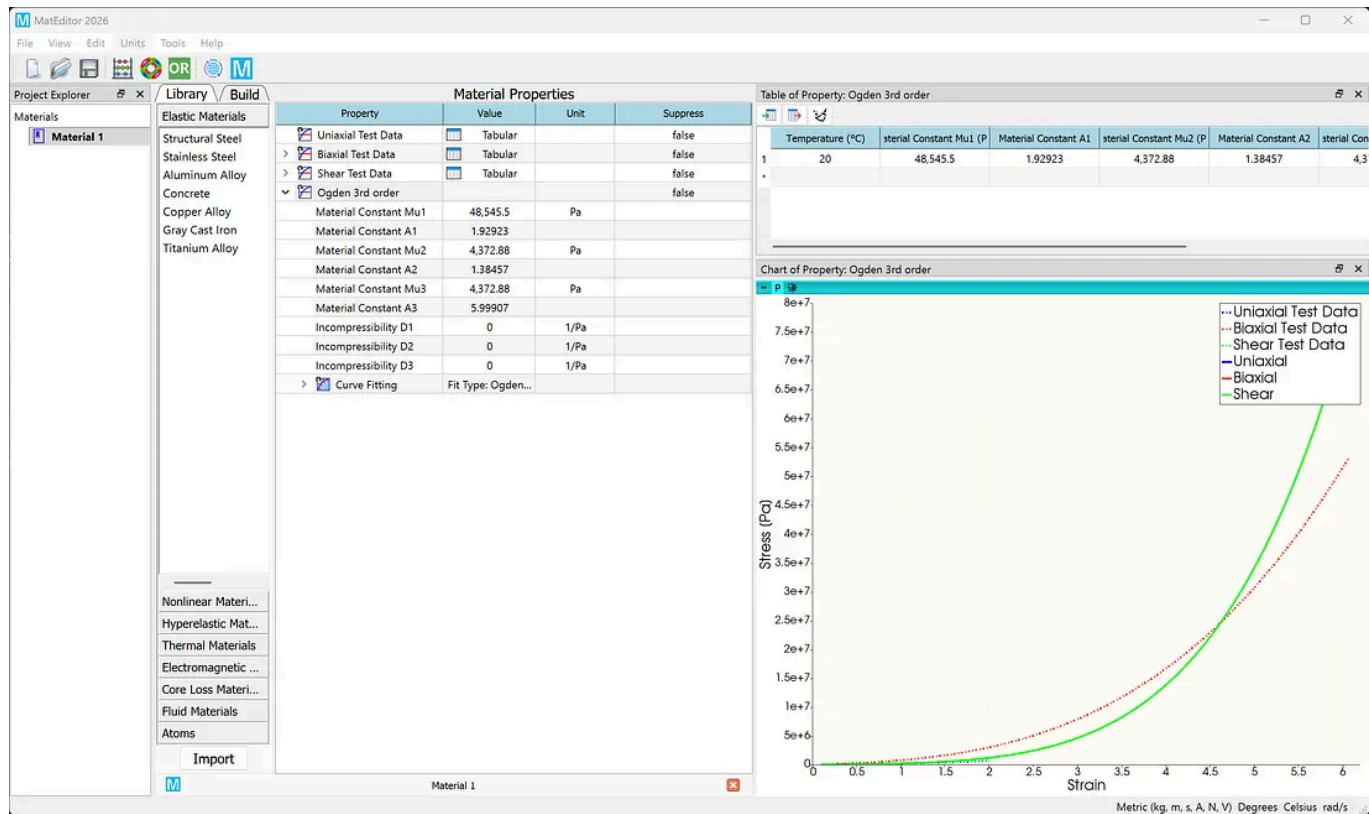
8. Ogden

The Ogden hyperelastic model is one of the most powerful and flexible hyperelastic models available today. It is constructed directly based on the principal stretches λ_i . The strain energy density function of the Ogden model adopts a series superposition form expanded by order n , where n represents the model order.

$$W = \sum_{i=1}^N \frac{\mu_i}{\alpha_i} \left(\bar{\lambda}_1^{\alpha_i} + \bar{\lambda}_2^{\alpha_i} + \bar{\lambda}_3^{\alpha_i} - 3 \right)$$

- 1st-order Ogden: 2 parameters, suitable for small deformations and simple rubber materials, highest computational efficiency.
- 2nd-order Ogden: 4 parameters, balances accuracy and efficiency, mainstream for general industrial simulations.
- 3rd-order Ogden: 6 parameters, suitable for large-deformation, highly nonlinear complex elastomers, highest fitting accuracy. The Ogden model maintains solid predictive capability even when strains reach 700% or higher.

The Ogden model has relatively more parameters, and higher-order versions require combined fitting of multiple sets of test data, such as uniaxial, pure shear, and equibiaxial, to prevent non-physical solutions and overfitting. Parameters must satisfy thermodynamic stability constraints, and reasonable bounds must be set during fitting. For example, set $\mu_1 > 0$, $\mu_2 < 0$, $\mu_3 > 0$, $1 < \alpha_1 < 10$, $-10 < \alpha_2 < 10$, $-15 < \alpha_3 < 15$.



Initial values for curve fitting are often set as follows: $\alpha_1=2$, $\alpha_2=2$, $\alpha_3=6$, $\mu_1=0.6\mu_0$, $\mu_2=0.1\mu_0$, $\mu_3=0.05\mu_0$, where the initial shear modulus $\mu_0=E/3$. Young's modulus E is obtained from the slope of the test data at low strains.

The empirical data fitted by common Ogden hyperelastic models is as follows:

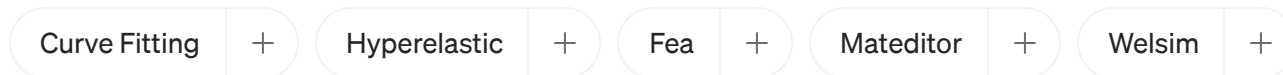
Material Type	Parameter	Typical Value Range	Description
Soft Rubber / Silicone	μ	0.01 ~ 0.5 MPa	Low modulus, gentle c
	α	-10 ~ 10	Usually includes posi
Industrial Hard Rubber	μ	1.0 ~ 10.0 MPa	High stiffness, large
	α	1.5 ~ 20	Large positive expone
Biological Soft Tissue	μ	0.001 ~ 0.1 MPa	Extremely soft, highl
	α	10 ~ 50	Exponents are usually

Conclusion

Although curve fitting has relatively mature numerical methods, fitting material parameters for finite element computation requires the consideration of many physical constraints, ensuring that the parameters are conducive to convergence in subsequent finite element calculations. Oftentimes due to the lack of test data, curve fitting for hyperelastic materials becomes complex which places high demand on both developers and users.

The main test data required for hyperelastic material model fitting includes uniaxial tension, equibiaxial tension, and pure shear deformation data. The author discusses this in detail in the article “[Hyperelastic Material Models and Curve Fitting](#)”.

Currently, WELSIM can accurately fit hyperelastic model curves. The independent and free engineering software MatEditor and CurveFitter also feature the aforementioned curve fitting capabilities. Users can download and use this software directly.



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